## The chain of command

If a national-scale project works and we build a superintelligence, prevent it from being stolen, and manage to contain and steer it, a few people will find themselves in control of a world-defining technology. The decisions those people make will shape the future of America and the world beyond. Even if this project achieves all its other objectives, it will fail if those in control use their power in a way that undermines American ideals.

The concentration of power that comes with superintelligence is as much a problem for the private AI labs who might build it as it would be for a national project. That's why frontier AI researchers, startup founders, and investors — including some of the most influential figures in Silicon Valley — have already started privately raising concerns about the leadership of some of those private AI labs. To quote one frontier researcher as he was discussing a particular top lab's CEO, "It seems clear he's a pathological liar, and I think there's a good chance that he's a full-on sociopath. He's power-hungry, and I'd bet he's willing to trade U.S. national security interests for a better chance of controlling superintelligence. As far as I know, he's not opposed to making compute clusters in [regions under totalitarian control]. He's probably up there in terms of one of the worst people to run a government AI project." He later added, "Very many people who've worked with him closely feel the same – maybe not as strongly, but basically the same." At the time, we were surprised enough by this position that we interviewed several other individuals with professional exposure to the CEO in question. And while their views did vary, a majority did generally agree with that researcher's assessment.

While we found that case surprising and notable, the issue isn't whether or not a specific AI executive is willing to compromise U.S. interests or is otherwise driven by a desire for personal power. The real problem is that all the checks and balances on power that American democracy depends on simply fall apart in the face of superintelligence.[57]

Many frontier lab insiders and top Silicon Valley investors openly expect that the capabilities of near-future AI systems will make it possible for one person to create mutli-billion dollar

[57] Several whistleblowers shared their concerns with us about current checks and balances at frontier labs. As one former OpenAI researcher told us, referring to OpenAI's new board, "Frankly, the function of the Board is to pretend to oversee the company." Of course opinions will differ, and we spoke to some current OpenAI employees who said they did not share this view. This level of oversight would also be perfectly fine at a random software company. But it's not strong enough for a company that's said it's aiming to build a superintelligence.

businesses alone. We think this is probably true, but its implications obviously don't stop at "single-employee unicorn startups". If AI really does provide this much leverage, it could easily make it possible for a single person at a computer console to launch nation-crippling cyberattacks or automated persuasion campaigns without anyone else signing off, or even knowing about it.

Power corrupts, and absolute power corrupts absolutely. This level of power is dangerous for a fundamental reason: it threatens basic American freedoms. The Founding Fathers wrote separation of powers into the constitution for a reason. The constitutional mechanisms they introduced have stood the test of time, but they weren't built to handle the extreme power that could come with technology like superintelligence.

Superintelligence will concentrate power like we've never seen before, and in ways that aren't necessarily obvious. If Xi Jinping had a superintelligent AI deployed at scale, he wouldn't need to keep his population under control by conventional means — or even necessarily to keep them alive at all: ultimately, his tax revenues would come from an automated economy, in which people are mostly dead weight. Even in America, a 4-year political appointee who happens to oversee a superintelligence project might end up wielding more power than any leader in history ever has, despite not having been appointed with anything close to that level of responsibility in mind. What's more, the person or group that controls superintelligence may be able to use it to circumvent their chain of command and democratic restraints in any number of ways, including by deceiving or persuading others that they're doing the right thing.

The imperative to prevent concentration of power and protect the project from corruption has to be traded off against two hard requirements. The first is time: checks and balances take time to develop and they slow down progress. And the second is security: information relevant to major decisions about the direction of the project will have to be closely held to prevent damaging leaks.

But too much concentration of power in a superintelligence project isn't something that can be fixed after the fact. The chain of command for a national AI project will need to have oversight over both the superintelligent AI itself, and also over the AI systems the project develops on the path to superintelligence, if they have specific superhuman capabilities like persuasion and deception that could compromise the project from the inside. We won't get many chances to get this right.

The chain of command needs to be put in place before it's needed, or the entire project could be corrupted by a small group of people. Thanks to automated AI research agents, frontier

labs' most advanced activities could ultimately be controlled by just a few people — or even just by one. In fact, one frontier researcher, who was citing OpenAI as an example, remarked "Man, at some point the silo might be literally [OpenAI President] Greg Brockman".

**Recommendations:** If a national project has perfect security *and* it manages to solve the problem of controlling a superintelligent AI, then a small handful of people will end up in control of the most powerful technology ever created. AI isn't like nuclear weapons, where most of the power comes from the technology's destructive potential. If you really have a superintelligent AI under your control, you might be able to reshape the world directly on your own. That's a recipe for the worst kind of corruption – especially because the demands of project security mean that the group in control needs to remain small.

This problem only matters if we can both secure and control a superintelligence first — if we fail at either of those things, then it won't matter how good our oversight is. But we still need to think about this now, because security, control, and oversight are problems that could come up in quick succession one after another. It could be a pretty short jump from an AI that's so capable that destroying or stealing it becomes a CCP strategic priority (security), to an AI that's dangerously hard to control (control), to an AI superweapon that centralizes power in a totally unprecedented way (chain of command).

Whatever form a push to superintelligence takes, **we'll need to have checks and balances in place from the very beginning that will prevent the individuals in charge from using the technology for their own personal or ideological reasons**. But just putting the right people in charge won't be enough. Not only does power corrupt, but we should be prepared for the possibility that AI systems developed on the path to superintelligence will develop the ability to persuade or compel their operators to do things they normally wouldn't.

The checks and balances we build will need to guide us through a transition period, when we have AIs that are very capable but still below the level of superintelligence. We also need to be able to halt development if we get security data that suggests a serious breach, technical data that indicates that controlling a superintelligence will be too difficult under current conditions, or indications that oversight mechanisms are at serious risk of failing. **And we'll have to build these checks into a project well before they're actually needed**.

No one has experience designing an oversight mechanism for such a speculative, high-stakes project. That said, the nuclear chain of command is a good example of a high-stakes context in which security, response times, and democratic accountability all have to be balanced carefully. We spoke to specialists in nuclear deterrence, security, and defense to figure how oversight

mechanisms for nuclear technology could inform the design of a superintelligence project, and were surprised at some potential quick wins that they flagged.

First, it's a common misconception that the nuclear chain of command always begins with the President. In some contexts, authority to launch nukes is delegated to base commanders or similar personnel to ensure short response times (though additional checks do apply to them). Similarly, in the context of a national superintelligence project, there will surely be game-time decisions that cyber operators or national security leadership will have to make quickly and with minimal intelligence signatures, but these can be narrowly scoped. Figuring out exactly which decisions can be made in real-time by a small set of people, and which need to be escalated to a decision-maker with greater democratic accountability is an obvious place to start.

Some other questions project designers will have to answer include:

- How many people should have to give independent sign-off for major decisions – and which decisions count as "major"?
- We'll need to test and verify the integrity of the project's chain of command – both the individuals within it, to detect potential signs of duress or foreign influence, and the communications channels they rely on. How should that be done, and by whom?
- What technical mechanisms, such as codes and keys, should be used? In the current paradigm, users provide AI systems with a prompt to give it instructions. That's quite different from the far more restricted action space available to the operators of nuclear weapons (roughly: where do I aim, and should I launch or not?). Presumably we want some mechanism to validate the prompts that are fed to the project's WMD-like models; how should that happen?
- The technology the project develops could have tremendous economic value. How can we ensure that there's an "offramp" that allows capabilities we develop in the project's secure environment to enter the free market to drive growth and prosperity?
- How do we maintain democratic accountability for project leadership? The project needs to be ultimately responsible to elected representatives of the popular will, the same way our military chain of command operates under civilian leadership.
- How do we design the right incentives for project leadership? What mechanisms can we use to remove participants if they seem to be compromised, and how can we design those mechanisms so that they can't themselves be compromised?
- The space between incentives and behavior is filled by culture. What is the appropriate culture of the project team? How should it balance the imperative to move fast with the

security, oversight, and control risks that the project could face? And what does that imply about personnel selection for project leadership, and more operational roles?

Apart from the basics, there are some easy wins we already know from experience that we'll need.[58] **For example, the oversight team will need to talk to the technical researchers on the project directly and constantly, instead of getting all their information through the leadership team.** Right now, some frontier AI labs use their "policy teams" as a smokescreen to filter the information they pass on to the government.[59] Government officials get fed pre-selected information from policy teams and lab leadership, when they should instead be talking directly to the researchers and engineers who are doing the work to get a true picture of what's going on (including those working on capabilities, security, and control). We've heard this complaint privately from dozens on both sides. Some government teams know they aren't getting the full picture but they aren't able to talk to the technical teams. And many technical researchers want to share their concerns with the government but their labs don't allow them to. That kind of obfuscation won't fly in a project like this one, and in fact we should put in place explicit whistleblower protections and a duty to report to make sure problems don't get swept under the rug.

Setting up good oversight will be hard. But the good news is that if we do this well, it will give us an advantage over adversaries in the AI race as well. If there's a global race to superintelligence, countries and people will have to choose sides. If this happens, the United States can give itself an advantage by publicly committing to credible, democratically accountable oversight of its project. A Chinese effort would be much less credible because democratic oversight of a project like this is incompatible with China's recent history, the CCP's culture, and its values. We should lean into this, commit to as much transparency and oversight as we can, and do it boldly and proactively. If Chinese messaging is the first to emphasize a democratic mandate — no matter how little real credibility those claims may have — we'll be seen as playing catch-up to China in a domain where our values should have placed us first. We should also commit to sharing the benefits of superintelligent AI to support our allies, and to encourage them to remain aligned with American superintelligence and national security efforts along the way.

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[58] There are also some easy wins that we don't have experience with yet but that we'll obviously need in the future. For example we'll need operational controls that scale with the capabilities of the AI — things like maintaining digital chains of custody, and preventing any project team member from querying the AI on their own if the AI's persuasive capabilities are rated high enough.

[59] This has been a recurring theme in our conversations with AI lab insiders.